MIL-L-52052D(ME)
9 September 1980
SUPERSEDING
MIL-L-52052C(ME)
29 May 1967

MILITARY SPECIFICATION

# LIFE PRESERVER, PARACHUTIST: TYPE B-7

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a life preserver with an oral and a mechanical inflation device.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-B-601 | -Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | -Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | -Boxes, Shipping, Fiberboard. |

FSC 4220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-1188 | - Commercial Packaging of Supplies and Equipment. |

DRAWINGS

ME

| | |
|---|---|
| TA 13215E6127 | - Life Preserver, Parachutist, Type B-7. |

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

| | |
|---|---|
| D413 | - Adhesion to Flexible Substrates. |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Description. The life preserver shall be in accordance with top assembly TA 13215E6127 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the life preserver. Any data (i.e., shop drawings, layouts, flow shuts, processing procedures etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request for inspection by the contracting officer or his designated representative.

3.2 First produced. The contractor shall furnish one or more life preserver for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce life preservers that comply with the requirements of this specification. Examinations and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government.



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3.3 Materials. Materials shall be as specified herein and as shown on applicable drawings. The materials, except for the metallic parts used in the construction of the life preserver, shall have been manufactured within 12 months prior to the date of delivery of the life preserver (see 6.3).

3.4 Performance.

3.4.1 Cemented joints. Surfaces to be joined by cementing shall be clean and free of all foreign matter prior to the application of adhesive. Adhesive shall be evenly applied to each surface and the surfaces joined. All entrapped air and wrinkles shall be removed by rolling or stretching. The minimum strength for cemented joints shall be 7.5 pounds per inch of width where coated side of cloth is joined to coated side and 5.0 pounds per inch of width where coated side of cloth is joined to uncoated side of cloth.

3.4.2 Mechanical inflation. A quick hard pull on the lanyard cord attached to each inflation valve shall cause the carbon dioxide cylinder to be punctured, and shall release carbon dioxide gas to inflate the flotation cell. Inflation and expansion of the cell shall cause the thread securing the packet to fail so that the packet will open and the cell inflate fully.

3.4.3 Oral inflation. Oral introduction of air through the oral introduction valve to a pressure of 1 psig shall fully inflate each flotation cell when cells are not secured inside the packet.

3.4.4 Pressure and leakage. Flotation cells shall withstand an internal air pressure of five psig for five minutes without material failure or separation of cemented joints, and shall maintain a final corrected internal air pressure of not less than 1-1/2 psig when tested as specified in 4.5.2.3. Flotation cells shall be free of holes, cuts, abrasions, or any imperfection or damage.

3.5 Workmanship. The life preserver shall be fabricated and assembled with correct mating and alignment of parts. Quality of workmanship in the fabrication of flotation cells and packets shall not allow these parts to become spotted or stained by the adhesive used in fabrication or by any foreign material such as dirt, grease, oil, or paint.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.



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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. The inspections shall be classified as follows:

(a) First produced inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection comparison (see 4.6).
(d) Inspection of packaging (see 4.7).

4.3 First produced inspection. Four life preservers will be selected at random by the Government from the first 20 life preservers being produced from production tooling, examined as specified in 4.5.1 and tested as specified in 4.5.2.2, 4.5.2.3, and 4.5.2.4 in that order to determine conformance to the requirements of this specification. The examination will be performed by the Government at a site selected by the Government; the tests will be performed by the contractor and witnessed by the Government at a site selected by the contractor. Acceptance of the first produced life preservers shall not exclude the remaining life preservers from the quality conformance and acceptance provisions specified in Section 4.

4.3.1 Inspection failure. Failure of a first produced life preserver to meet any requirement specified herein during and as a result of the examination and tests specified in 4.3 shall be cause for rejection of the first produced life preservers and shall be cause for refusal by the Government to continue acceptance of production of life preservers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accompolished by the contractor at no cost to the Government on life preservers previously accepted and produced under the contract. Any deficiencies found as a result of the first produced testing will be considered prima facie evidence that all life preservers accepted prior to the completion of first produced testing and similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Life preserver. Sampling for examination and tests of complete life preservers shall be in accordance with MIL-STD-105, inspection level S-3.

4.4.1.2 Adhesion samples. Five representative samples of the coated side of the cloth cemented to the coated side of five representative samples of the coated side of the cloth cemented to the cloth side shall be prepared during manufacture of flotation cells for each lot of life preservers. The same cement and techniques used in the preparation of cemented joints of flotation cells shall be used in the preparation of the samples.



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4.4.2 Examination.

4.4.2.1 Individual. Each life preserver shall be examined for the critical defects specified in 4.5.1. Presence of a critical defect shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 6.5 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Individual. Each flotation cell shall be tested as specified in 4.5.2.3. Failure of the test shall be cause for rejection.

4.4.3.2 Samples.

4.4.3.2.1 Life preserver. Samples selected in accordance with 4.4.1.1 shall be tested as specified in 4.5.2.2. Failure of test shall be cause for rejection of the entire lot.

4.4.3.2.2 Adhesion samples. Adhesion samples prepared in accordance with 4.4.1.2 shall be tested as specified in 4.5.2.5. Failure of test shall be cause for rejection of the entire lot of life preservers.

4.5 Inspection procedures.

4.5.1 Examination. Life preservers shall be examined for the following defects:

Critical

1. Flotation cell material not as specified.
2. Flotation cell cemented joints not as specified.
3. Holes, tears, cuts, abrasions or any imperfections or damage to flotation cells.
4. Valves not as specified on the drawings.
5. Carbon dioxide cylinder not as specified on the drawings.
6. Packing of flotation cells in packets not as specified on the drawings.

Major

101. Material other than flotation cells not as specified.
102. Dimensions not as specified on the drawings.
103. Components missing or not as specified on the drawings.
104. Assemblies incomplete or not as shown on drawings.
105. Stitching missing or not as specified on the drawings.
106. Bartacks missing or not as specified on the drawings.
107. Edges of nylon fibrous materials not seared as specified in the drawings.



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Minor

201. Identification markings missing or not as specified on the drawings.
202. Color not as specified on the drawings.
203. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Life preservers which have passed the test specified in 4.5.2.2 shall have the spent carbon dioxide cylinders replaced and shall be reassembled in the packet. These life preservers shall be subject to reinspection as specified in 4.4.

4.5.2.2 Operation. The life preserver shall be fitted to an individual or mannequin and the lanyard cord on each valve shall be given a hard pull to puncture the carbon dioxide cylinder and inflate the flotation cells. Failure of either flotation cell to inflate fully shall constitute failure of this test.

4.5.2.3 Pressure and leakage test.

4.5.2.3.1 Test conditions. Atmospheric pressure for the duration of the test shall be not less than 28 or more than 32 inches of mercury; the ambient temperature shall be between 59° F and 95° F., and the relative humidity shall not exceed 80 percent.

4.5.2.3.2 Test. Each flotation cell shall be inflated by introducing moisture free air through the oval inflation valve to a minimum pressure of 5 psig and that pressure maintained for not less than 5 minutes. The pressure shall than be reduced to 2 psig and the cells allowed to stand for a minimum of 15 minutes. The pressure shall be readjusted to 2 psig by adding or bleeding air as required and the atmospheric pressure and ambient temperature noted. After a period of not less than 24 hours, the internal pressure of the flotation cell shall be measured and the atmospheric pressure and ambient temperature noted. Any change in atmospheric pressure or ambient temperature at start and completion of test shall be determined and the final internal pressure of flotation cell shall be corrected as follows:

(a) Temperature correction. For each degree Fahrenheit rise in ambient temperature subtract 0.031 psi. For each degree Fahrenheit drop in ambient temperature add 0.031 psi.

(b) Pressure correction. For each 0.1 inch of mercury rise in atmospheric pressure add 0.049 psi. For each 0.1 inch of mercury drop in atmospheric pressure subtract 0.049 psi.

Nonconformance to 3.4.4 shall constitute failure of this test.



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4.5.2.4 Cemented joints. Ten samples shall be taken at random from each flotation cell in each sample life preserver, and tested in accordance with ASTM D413. Five of the ten samples shall be of coated side cloth cemented to coated side of cloth. The other five samples shall be of the cloth side of cloth cemented to coated side of cloth. Failure of one or more samples to meet the minimum strength requirement specified in 3.4.1 shall constitute failure of this test.

4.5.2.5 Adhesion. Adhesion samples shall be tested in accordance with ASTM D413, Machine Method. Failure of any test sample to meet minimum strength requirements specified in 3.4.1 or the failure of the supplier to furnish adhesion samples as specified in 4.4.1.2 shall constitute failure of this test.

4.6 Inspection comparison. The Government may select life preservers at any time during the contract production period and subject these life preservers to the examination specified in 4.4.2 and the test specified in 4.5.2.2, 4.5.2.3 and 4.5.2.4 to determine that the quality of the selected life preservers is equal to the quality standards established during initial production test. The inspection will be performed by the Government at a site selected by the Government. Life preservers will be selected at random from those which have been accepted by the Government.

4.6.1 Inspection failure. Failure of any life preserver when inspected as specified in 4.6 may be considered cause for refusal by the Government to continue acceptance of life preservers until objective evidence furnished by the contractor reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.7 Inspection of packaging.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

| | |
|---|---|
| 108. | Materials and methods not as specified for level A or B. Each incorrect material and each incorrect method shall constitute one defect. |
| 109. | Each life preserver not bundled as specified for level A. |
| 110. | Each box not closed and waterproof sealed as specified for level A. |
| 111. | Gross weight of wood boxes exceeds 200 pounds for level A. |



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112. Fiber boxes exceeding the size and weight limitations for level B packing.
113. Fiber boxes not waterproof sealed and reinforced with bands of tape as specified for level B packing.
114. Marking illegible, incomplete, incorrect or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each life preserver shall be bundled by placing the two flotation cell packets together with their flap (opening) sides adjacent. Web straps shall be wrapped snugly around the longest circumference of the bundled packets, avoiding unnecessary twisting of the straps. Each bundled life preserver shall be placed in a close-fitting fiberboard box conforming to PPP-B-636, W5c, style optional. Each box shall be closed and waterproof sealed with tape in accordance with Method V of the appendix in the box specification.

5.1.2 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A. The life preservers, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style optional, or PPP-B-621, class 2, style optional. The gross weight of each wood box shall not exceed 200 pounds. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification.

5.2.2 Level B. The life preservers preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, V3c, style optional, or as specified for level A, using domestic class boxes. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. The life preservers, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract, marking shall be in accordance with MIL-STD-129 for military levels of protection. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The life preserver is intended for use as emergency equipment by paratroopers participating in over-water flights.



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(a) Title, number and date of this specification.
(b) Time frame required for submission of first produced (see 3.2).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.4 Revision changes. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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